



**SCHOOL OF ELECTRICAL ENGINEERING
AND TELECOMMUNICATIONS**

Thesis B Progress Report

by

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Contents

Contents	2
Abbreviations	3
1 Progress Report	5
1.1 Thesis A Summary	5
1.2 Single Module Design	5
1.3 EMI Fault Diagnosis and Resolution	6
1.4 Power Stage Testing Results	9
1.5 Design of Final Implementation	11
2 Reflection	13
3 Revised Plan	15
3.1 Revised Project Plan	15
3.2 Risk Management Strategy	16
Appendix	17

Abbreviations

ADC Analog to Digital Converter

CAN Controller Area Network

FD-CAN Flexible Data-rate CAN

COTS Commercial Off-The-Shelf

EL Electroluminescence

EMI Electromagnetic Interference

FET Field-Effect Transistor

GaN Gallium Nitride

GND Ground

GUI Graphical User Interface

HRTIM High-Resolution Timer

HV High Voltage

I2C Inter-Integrated Circuit

IC Integrated Circuit

ISR Interrupt Service Routine

MCU Microcontroller Unit

MISO Multiple-Input Single-Output

MPPT Maximum Power Point Tracking

NTC Negative Temperature Coefficient

OCP Over-Current Protection

PCB Printed Circuit Board

PCBA PCB Assembly

PV Photovoltaic

SW Switch node

UART Universal Asynchronous Receiver-Transmitter

USB Universal Serial Bus

UVLO Under-Voltage Lockout

Chapter 1

Progress Report

1.1 Thesis A Summary

The goal of this thesis is to design a multi-channel MPPT solution for the Sunswift 8 solar vehicle. Given the unique automotive operating environment, understanding the requirements was critical to creating an optimal design. For this reason, the primary objective of Thesis A was to understand the requirements, such as bus voltage, panel V/I characteristics, communication interfaces, and other automotive related requirements. This was discussed thoroughly in the thesis A report. Furthermore, in thesis A, I took the time to understand loss mechanisms in synchronous switching converters, MPPT algorithms, and environmental factors leading to PV perturbations. Using this knowledge, I approached thesis B with an understanding of the design problem, allowing me to begin the design of a single module testing system.

1.2 Single Module Design

After the submission of the thesis A report, I started to develop a solution to characterise my chosen inductor, GaN switching device, and capacitors. Alongside this, I procured an STM32H743 Nucleo and designed a JST-GH adaptor board, allowing a single microcontroller to cleanly attach to 5 of these single modules. This setup is pictured in figure 1.1.

This design integrates two digital INA228 power sensors on the input and output for performing MPPT algorithms and efficiency analysis, alongside a HI/LI interface connected to the HRTIM channel on the STM32H743. It used a $33\mu H$ SER2211 inductor, a LMG2100R026

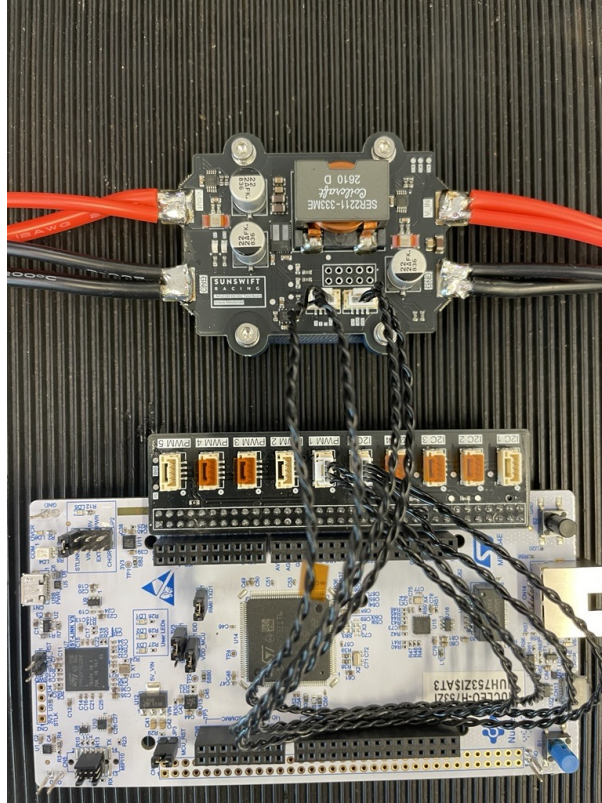


Figure 1.1: Single module testing setup

GaN half-bridge with an integrated gate-driver, and around $49\mu F$ of output capacitance. The intent was to use this design to de-risk making a large single-board implementation with 5 modules and to uncover any critical design flaws. Originally, the plan was to also implement a subsection of the final firmware on this microcontroller with 5 modules, but this did not come to fruition due to issues discussed in later sections.

1.3 EMI Fault Diagnosis and Resolution

After receiving this board, I developed a simple UART abstraction interface to allow a GUI to manipulate the HI/LI duty cycle, pictured in figure 1.2. This used a simple list of predefined serial headers that contained a single byte command ID, a single byte command size, and then the data package.

Following this, I uncovered the first design mistake. I had accidentally connected the HI/LI outputs of the HRTIM channels to the 'HI' and 'LI' inputs of the gate driver. However, in a synchronous boost converter, the 'LI' or low-side FET is the dominant controlling switch, and

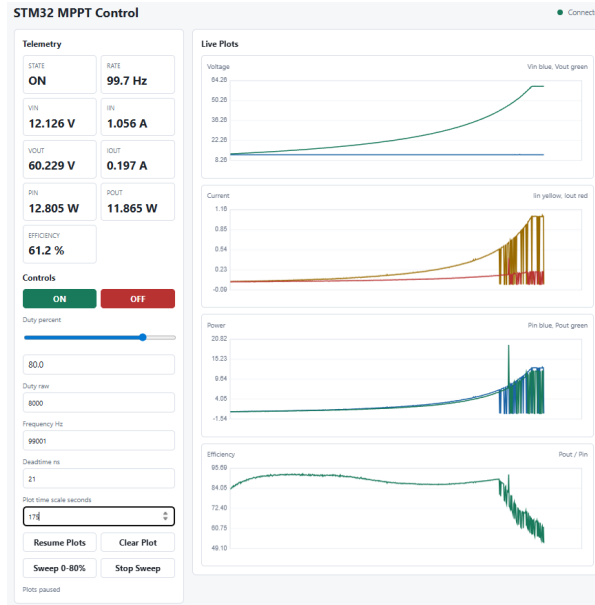


Figure 1.2: GUI for single module testing

the high-side FET is the synchronous diode replacement. Since I made this assumption when implementing the timers, I caused the output duty cycle to start at 100%, making the output voltage extremely high when starting. Thankfully, this did not cause any issues and was able to be rectified by depinning the JST-GH cables and repinning them correctly.

The next major issue with this board is evident in figure 1.2. When the current begins to increase, the digital power sensors start to rapidly become noisy and even drop-out. This happened more noticeably at higher switching frequencies, and higher power throughput. This means that the issues was very likely EMI related, due to the $\frac{di}{dt}$ increasing as f_{sw} increases as I_{out} increases. Large $\frac{di}{dt}$ pulses inject noise into the system by injecting voltage through parasitic coupling inductance.

My first instinct was to probe the I2C bus, and the system ground, to inspect for noise above the V_H threshold of the I2C signals, and for ground bounces. This showed noticeable noise on the bus with a frequency matching f_{sw} , although the ground seemed relatively unaffected. However, I noticed that when making the inductor footprint, although I added a keepout directly under the inductor, this did not extend to the edges of the footprint like it should have. This meant that I had inadvertently routed the I2C bus directly under the edge of the inductor. To rectify this, traces were cut and rewired around the edge of the board as seen in 1.3. This did

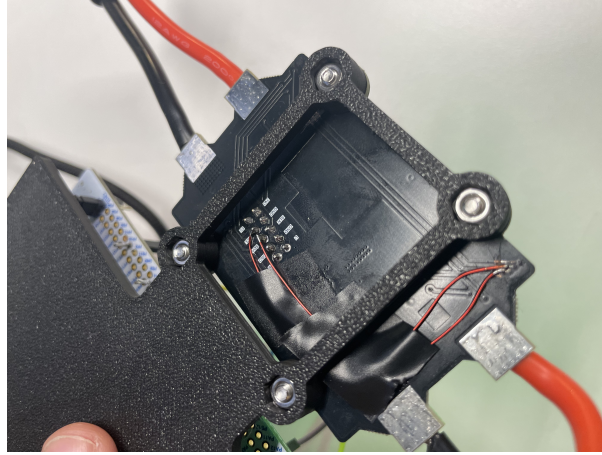


Figure 1.3: Rerouting I2C traces around inductor

noticeably help and increased the threshold at which I2C sensor dropouts were occurring, but did not fix the problem.

Following this issue, I then discussed with my supervisor, John Fletcher, to determine what else could be causing these dropouts. He worked with me to help determine that there could be a number of factors. We measured the ripple current of the inductor using the input shunt, and at a nominal input of 8.22A, we found the ripple was as high as $\Delta I \approx 4A$, saturating the inductor every cycle. This meant that, due to the placement of the input current sensor (which was the worst offender), magnetic flux leakage could be causing issues inside the integrated circuit. This was verified by rotating the inductor position to ensure the flux leakage path did not intercept the chip. However, this did not fix the issue.

At this point, I had paused fixing this issue to avoid delaying this thesis and began to manually collect results that are discussed in the next section. However, I knew this issue had to be solved to confidently move to the final implementation. Since I knew it was an EMI related issue, I began looking harder into the layout of the board. I then realised that although I had rectified and checked the positive current path out of the I2C sensors (towards the microcontroller), I had neglected to check the return current path. I then realized that I had forgotten to stitch the ground plane surrounding the INA sensors, forcing the return current path right through the SW node of the FET (effectively, the current path was using the thermal GND vias of the FET). This return path is seen in figure 1.4. This would explain why the bottom INA228 sensor was seeing more dropouts than the top, since the return current path is also routed under

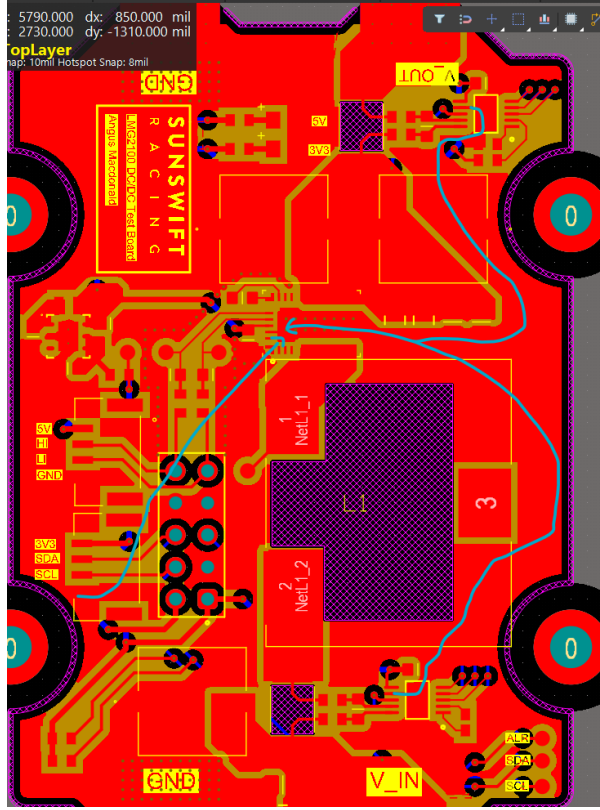


Figure 1.4: Return current path illustrated on layout

the inductor. To test this hypothesis, I cut the ground plane in multiple locations and soldered wires directly from the chips to the left-side of the board that had a more direct path to the microcontroller, pictured in figure 1.5. This solved all dropout issues and gave me the confidence to move to the final hardware implementation.

1.4 Power Stage Testing Results

Whilst I was still resolving the EMI issue present in these single module test boards, I was still able to manually collect data using the bench supply and load. Both of these devices are able to measure the supplied voltage and current to the nearest mV and mA respectively. This test setup is pictured in figure 1.6.

Using voltage sense wires, I was able to bypass the effects of voltage drop on the connecting cables, and extract some positive results that validated this converter approach. For the first test, I collected efficiency data at a f_{sw} of $300kHz$ with a constant duty cycle. The results are shown in figure 2. It can be seen that the peak efficiency in this basic switching configuration is

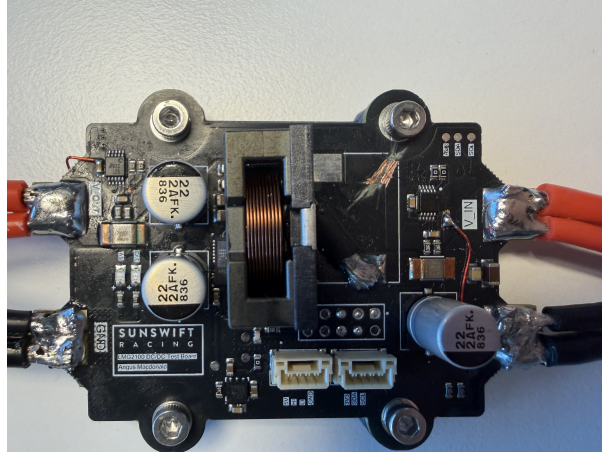


Figure 1.5: Return current path rework



Figure 1.6: Single module testing setup

about 98%, which is a good result and gives confidence with this converter design for the final implementation.

I also collected data for a frequency sweep, keeping a constant current of 7A and a constant duty cycle. This gave a peak efficiency of about 98.4% at 200kHz, although the current ripple at this frequency may be unacceptable for the MPPT controller to perform accurately. The results for this test are found in figure 3.

In comparison to my Thesis A prediction of 96.5% based on calculated switching and conduction losses, these results are higher. This is possibly due to having a shorter rise and fall time than expected on the FET's, leading to reduced switching losses. It is important to note that due to measurement error in the supply and load, as well as shunt tolerance, the peak efficiency measurement must be taken as an estimate rather than an absolute value.

After completing testing and debugging for the single module test boards, I was happy to move to the final hardware implementation. The rest of thesis B, up until the writing of this report, has focused on this design.

1.5 Design of Final Implementation

The design of the final implementation has undergone many schematic changes from thesis A and is at the point of schematic finalisation and initial layout. The Altium project files can be found in [github](#).

Major changes include the addition of hardware functionality for cycle-by-cycle inductor current control and overcurrent limiting. Using the INA310A2 and the input shunt, this integrated circuit allows for an analog current sensor output alongside a comparator output. Programming the comparator output for 12A allows for fault protection against large switched current pulses. This was deemed necessary due to the fairly slow, 100μs update rate of the INA228. This sampling rate is perfectly fine for slow MPPT algorithms, but due to the inductance and operating voltage, it could lead to large increases in current before the next detection point. Since $di = \frac{V}{L}dt$, in 100μs, a current spike of about 82A could occur if issues arise downstream. Al-

though this issue assumes that the voltage on the input will remain unchanged for $100\mu s$, which is unlikely given the IV nature of the input and the limited input capacitance. Additionally, the downstream components can handle this pulsed current, but they may increase degradation on the output aluminium electrolytic capacitors.

Another major change is swapping to a new VER2923 inductor that saturates at a higher current due to the expected ripple current this design will be tolerating. This new inductor has a 20% decreased inductance at about 9.2A, allowing for increased saturation headroom and hopefully a more efficient design.

This schematic has now also implemented an ideal diode on the output to the auxiliary battery bus to prevent rapid charging of the boost converter output capacitance when the battery is plugged in. This will also provide efficient reverse polarity protection and fault protection if the boost stages fail. An additional switch was also added to allow bypass of the ideal diode at the request of the Sunswift solar team to inject current into the panels for imaging.

Additionally, all auxiliary components have been implemented for the microcontroller, such as 5V and 3.3V power supplies, alongside communication interfaces for FD-CAN, USB, and programming. Schematics are available in the appendix.

Layout has also begun and is in progress with a 6-layer board. A 6-layer stackup of signal, ground, signal, power, ground, signal was chosen due to the considerable EMI issues faced on a signal board and the mixed analog/digital design. Some screenshots of the current layout are also provided in the appendix.

Chapter 2

Reflection

This thesis project has been a major learning experience. Beginning this thesis, I knew that taking a complex design from determining requirements to a finished product wouldn't be easy, but in practice, it has been far more time consuming than I originally thought. Over the course of thesis A and B, I have experienced feelings of extreme frustration and setbacks that I would have never expected. However, this has only made me more eager to find issues and solve the problem I set out to achieve at the beginning of this year.

As I have been progressing through my thesis, I have found that learning more about the project and the background has led me to believe I have made suboptimal design decisions. Being able to adapt to changes in learning is critical. A good example of this is finding out that having no cycle-by-cycle protection is a critical design flaw, and I needed to work out how to implement this into my existing design.

Having complete ownership of a complex project is unlike anything I have experienced at university or my time spent working in the aerospace/power electronics industry. At university, projects and problems are usually set out in a very black and white way, almost never getting the student to determine the requirements and figure everything out completely on their own. I have enjoyed going through this process and having the independence to make and reason critical design decisions. In comparison to my experience in the industry, where I have taken on similar smaller-scale projects, I have had far larger feedback and oversight into my design decisions. I found that I struggle when working individually, which leads to a lack of thorough checking and reliance on preexisting processes to catch errors and critical decisions early. The

best example of this is the issue I found in my testing boards, which would have been solved with two stitching vias at the decoupling capacitors, a common aspect that a review checklist would pick up on if I had gone through that process. I have now learnt that reviewing with a proper predetermined checklist will be necessary for the final implementation and all future hardware designs that I undertake.

The most valuable lesson I have learned so far is dealing with frustration and thinking clearly when nothing is working and you feel like you have tried everything. Sometimes taking a step back, and going to the fundamentals can help clear your mind and allow you to tackle the problem. An important lesson I have learnt from my supervisor is that you should never give up until you are confident you know what the problem is and how to fix it, and that in engineering every problem you find has a root cause that can be determined.

Moving forward, I will take these valuable lessons I have learned for a smoother and more successful experience in the remainder of thesis B and C, making sure I do not rush and skip steps, and take the time to critically review my work as I go.

Chapter 3

Revised Plan

3.1 Revised Project Plan

The updated project plan can be seen in figure 3.1. This plan contains a few noticeable differences from the original timeline. The first is I have de-scoped the project to not include a HV bridge converter. Originally, I thought the project was going to be simpler then what I have found, so I included this section to increase complexity. However, I have now found that this thesis project is already becoming very complex and hard to manage, so this section will be replaced in the Sunswift car with a COTS component. The other major difference is that due to issues with the test board, the project itself has been delayed by about 1 month when referencing the original plan. This is acceptable since the layout for the hardware implementation is almost done, and the project has been de-scoped. In thesis B, I intend to finish the final implementation hardware in the coming weeks, ideally by the end of week 6. I estimate manufacturing and PCBA will take about 2 weeks, so I should have a board by the end of week 8. The final two weeks of term will be spent on doing bring up testing and characterising boost converter efficiency and the function of all auxiliary circuits.

Moving into the holidays and thesis C, I will begin writing the firmware, which I estimate will be a difficult process and require considerable learning, given the fact that I have only

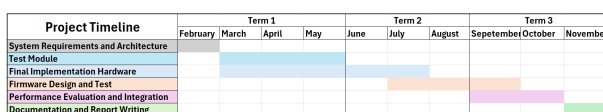


Figure 3.1: Revised project plan

written simple drivers and state machines in the past. Managing multiple control loops, handling many ISR's, and keeping the system safe and robust will be a major challenge that will take most of my holidays and the early weeks of term 3. I will also conduct comprehensive testing campaigns alongside this to evaluate hardware and firmware performance. The remainder of the term will be spent on documentation and report writing.

3.2 Risk Management Strategy

There are several major risks that I have to consider that could derail the revised timeline. The first risk is a re-spin in the final board. This could delay the project by over a month, so appropriate precautions will be taken. This includes conservative schematic and layout choices (such as a 6-layer board when it may be possible to route in a 4-layer board), jumpers on all nets I am unsure about, and understanding failure modes and potential ways to fix them without a re-spin.

The second major risk is inadequate resources to conduct the intended control approach with a switched current controller inside an MPPT controller for each channel. I have included the ability to not use this current controller and instead use a far simpler direct MPPT approach with hardware OCP.

The final major risk is rushing through the plan without proper due diligence. This leaves me more prone to making major mistakes and completely derailing the project. This will be mitigated with proper review processes involving multiple external sources such as Sunswift members, my thesis supervisor, and potentially my co-workers.

Appendix

This appendix contains schematic progress and current layout progress. If a better resolution is needed, please clone the [github](#) repository. Optionally, there is a PDF in the top-level directory that is available for download.

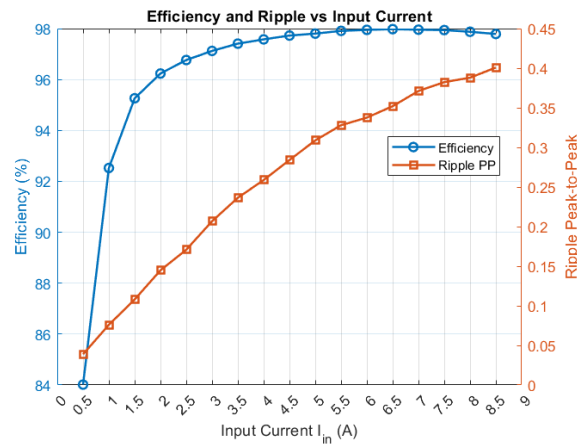


Figure 2: Efficiency vs current curve for single module testing

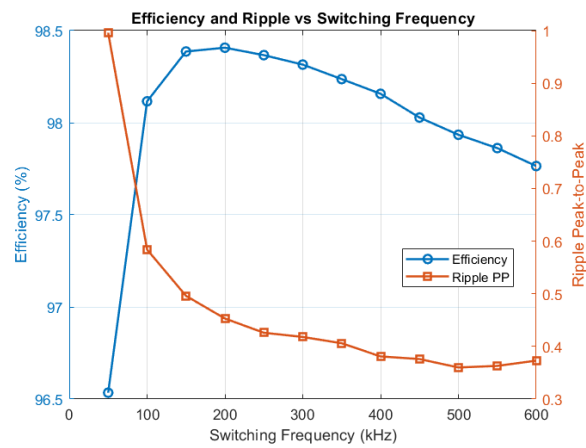


Figure 3: Efficiency vs frequency curve for single module testing

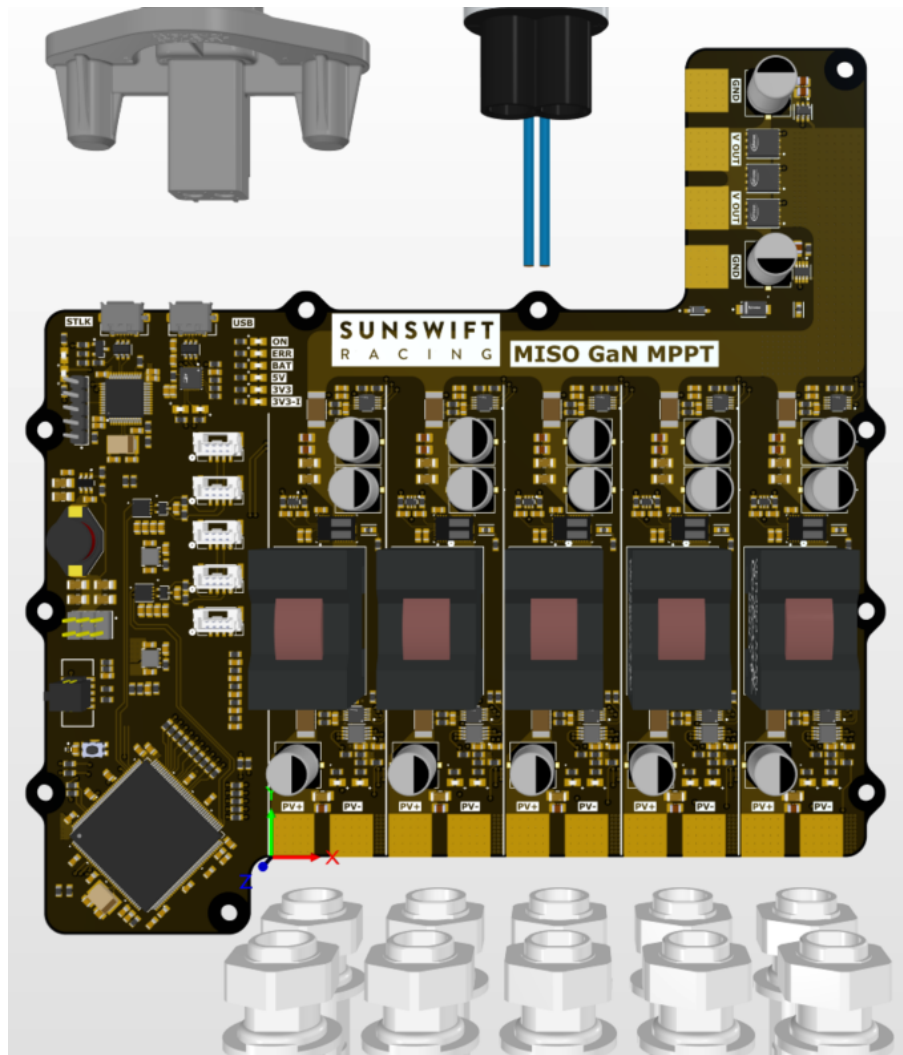


Figure 4: 3D view of current design

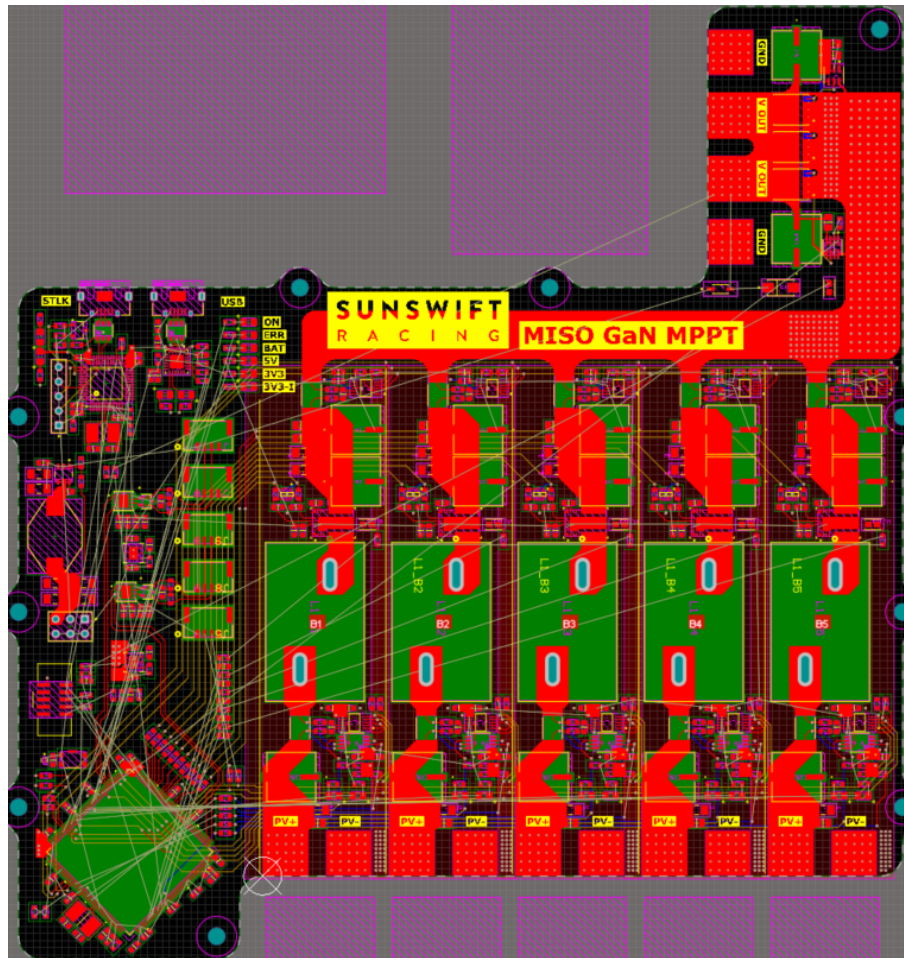


Figure 5: Current layout progress

